

EMPLOYMENT

LexisNexis Legal & Professional

Senior Data Scientist II

Raleigh, NC

Oct 2026 - Present

- Benchmarked 15 open-source structured decision models (System One) against four LLM baselines across six classification tasks; fine-tuned three Pareto-optimal models (best-model held-out macro-F1: 92.7%) to improve latency (M4 Pro, median: 177 ms, p95: 254 ms).

Senior Data Scientist I

May 2024 - Oct 2026

- Technical lead for an agentic legal drafting system; led its migration from RAG and continue to own development.
- On a 3-person team, increased the share of drafts rated suitable for users by attorney reviewers from 40% to 86% for complaints and 59.2% to 76% for motions.
- Reduced customer-facing product runtime by 24.8% through LLM call profiling and parallelization.
- Built a contrastive-learning filter using InfoNCE with alignment and uniformity losses to remove unrelated legal passages from search results; median filtering latency of 12 ms across 40 queries.
- Built team developer tooling: a Rust MITM proxy with request replay, a web UI for visualizing intermediate agent steps, and custom Python lint rules beyond `ruff`'s rule set.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Profiled high pod memory usage and introduced lazy module loading, reducing time at peak memory from 86.4% to 68% of runtime.
- Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

North Carolina State University

PhD Student

Raleigh, NC

Aug 2019 - Jul 2024

- Hyper-parameter optimization:** Proposed an optimizer that outperformed prior methods and ran 36.4% faster than Hyperopt on datasets used in the NeurIPS 2020 BBO Challenge.
- Defect prediction:** Proposed GHOST, an adaptive oversampling and hyper-parameter tuning pipeline that outperformed prior deep-learning methods on 14 of 20 defect-prediction datasets.
- Semi-supervised learning:** Developed GHOST2 for identifying actionable static-analysis warnings; achieved median AUC of 0.92 across eight open-source Java projects using 10% of the training labels.
- Teaching:** TA for 830 students over 5 semesters across CSC 230 (C and Software Tools), CSC 510 (Software Engineering), and CSC 591/791 (Automated Software Engineering)

RELEVANT PROJECTS

pi-permission-system (maintained fork)

TypeScript, Node.js

May 2026 - Present

[GitHub](#)

Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling.

Q&A System for Zotero

Rust, SQLite, LanceDB

Oct 2024 - Present

[GitHub](#) :: [crates.io](#)

Built an academic question-answering and semantic search system over a Zotero library. Developed a Rust RAG library with configurable LLM, embedding, and reranking providers, tool calling, and limits on concurrent embedding requests. Wrote a custom PDF parser for academic papers (median throughput $\approx$ 2,400 pages/s across 11 PDFs; M4 Pro).

RAISE

Python, Keras, PyTorch

Aug 2020 - Jun 2026

[GitHub](#) :: [PyPI](#)

Sole developer of a Python package for reproducible ML experiments, providing data loading, evaluation metrics, statistical tests, and reference implementations of research algorithms. Uses Keras with a PyTorch backend; tested on Python 3.11-3.14 across Linux, Windows, and Apple Silicon macOS. Used by multiple research groups; downloaded 79k+ times on PyPI.

Flatness-guided HPO

Python, PyTorch, Keras

Nov 2023 - May 2026

[GitHub](#) :: [Paper](#)

Developed a hyper-parameter optimizer using closed-form loss-curvature estimates to select configurations with flatter losses. Implemented comparative benchmarks on MNIST, SVHN, OpenML, and Bayesmark.

GHOST

Python, Keras

Developed the GHOST reference implementation for software defect prediction, combining adaptive oversampling with DODGE hyper-parameter optimization. Implemented configurable ablation experiments and Scott-Knott statistical comparisons using RAISE.

Jun 2020 - Mar 2022

[GitHub](#) :: [Paper](#)

SKILLS

Languages: Rust, Python, TypeScript, Java, C++, Gleam

ML/AI: Keras, PyTorch, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK

Frameworks: React, Flask, Litestar, FastAPI

Databases: LanceDB, MySQL, MongoDB, DynamoDB

Cloud: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)

EDUCATION

North Carolina State University

Ph.D. Computer Science

Raleigh, NC

Aug 2019 - Jul 2024

◦ Advisor: *Dr. Tim Menzies*

◦ Dissertation: [Guidelines for the Application of Neural Technologies in Software Analytics \(or: How to Do More with Less in SE\)](#)

PES University

B.E. Computer Science – Advisor: *Dr. Snehanshu Saha*

Bangalore, India

Aug 2015 - May 2019

HONORS

Distinguished Reviewer, ICSE 2026

Mar 2026

Google Developer Expert - ML and GCP

Mar 2025 - Present

Google Cloud Champion Innovator - Cloud AI/ML

Oct 2022 - Mar 2025

Google Cloud Research Innovator

Feb 2022

RECENT PUBLICATIONS

See full list on [Google Scholar](#).

1. Nagarsekar, A., Bhat, M. P. A., Nesarkar, A., Godhwani, V., **Yedida, R.**, Challa, A., Danda, S., & Saha, S. (2026). A Generalisation Signal Need Not Be a Model-Selection Signal. In *I Can't Believe It's Not Better: Failure Modes of AI in Biology Workshop (NeurIPS 2026)*.
2. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering*, 51(6).
3. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM*, 66(2), 75-81.
4. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering*, 49(4), 2856-2872.
5. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications*, 219, 119673.